



Complex Numbers: Equations to Quartics 3i

Subject & topics: Maths Stage & difficulty: Further A P1

Part A

$z^4 = 16$

Find the roots of the equation $z^4 = 16$.

+ i

Part B

Another quartic

Hence, using your findings from part A, solve the equation $w^4 = 16(1 - w)^4$. Give your solutions as decimals, rounding to 3 sf where necessary.

+ i

Adapted with permission from UCLES, A Level, Jan 2010, Paper 4727, Question 4.

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Argand Diagrams: Solving Inequalities 3i

Subject & topics: Maths Stage & difficulty: Further A P1

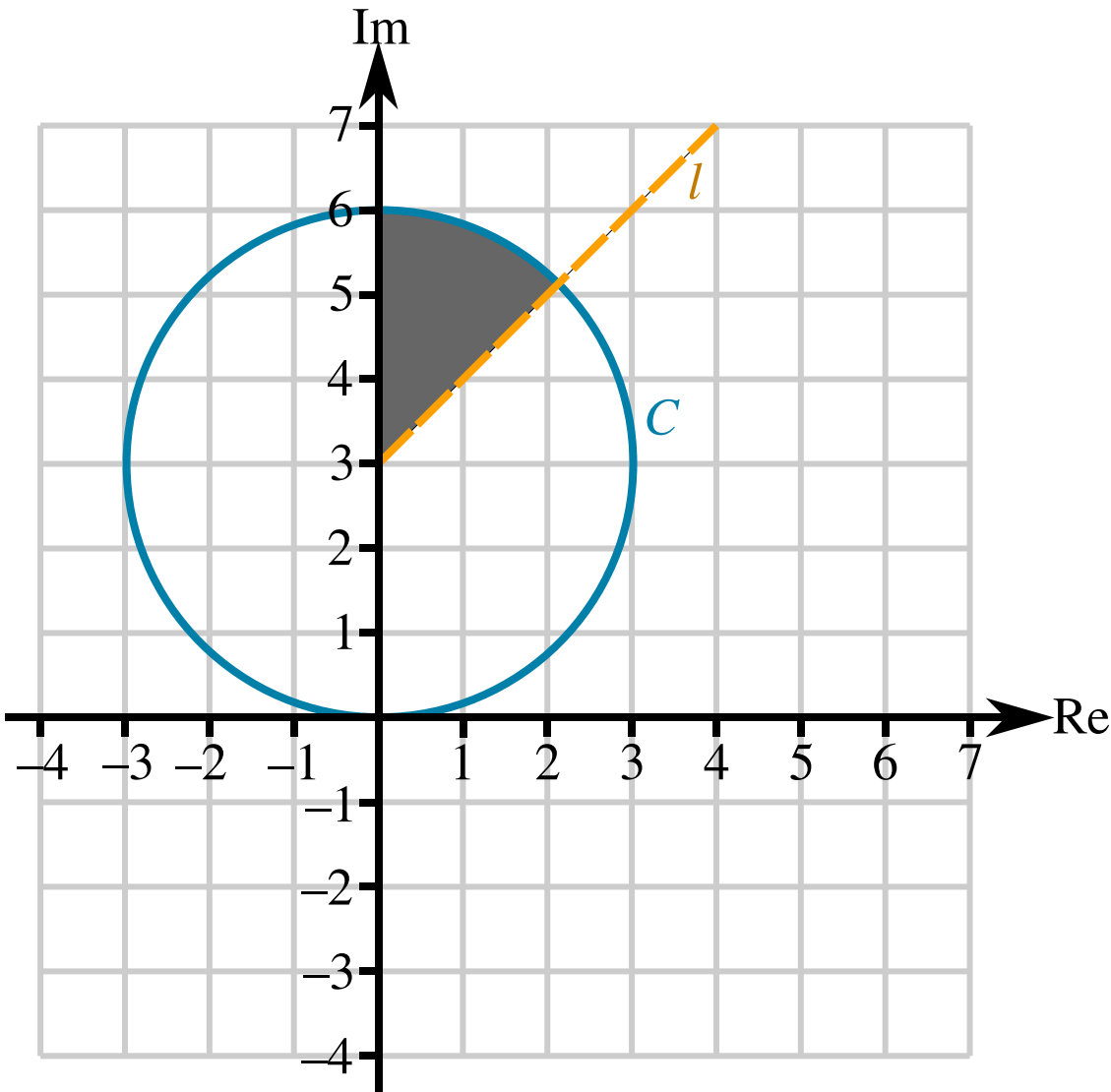


Figure 1: C and l are shown on a single Argand diagram.

The Argand diagram above shows a circle C and a half-line l . The circle has centre $3i$ and passes through the origin.

Part A

Equation of C

Give the equation of C in the form

$$|z - z_1| = a$$

where z_1 is complex and a is real.

The following symbols may be useful: i , z

Part B

Equation of l

The equation of l can be written in the form

$$\arg(z - \boxed{}) = \boxed{}$$

Use the items to complete the equation of l .

Items:

- −6

−3

0

3

6

−6i

−3i

3i

6i

− $\frac{\pi}{2}$

− $\frac{\pi}{3}$

− $\frac{\pi}{4}$

− $\frac{\pi}{6}$

$\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{\pi}{2}$

Part C

Inequalities

The shaded region includes its boundaries and is defined by two inequalities. They are in the form

$|z -$

$|$

and

$\leq \arg(z -$

$) \leq$

Use the items to complete the inequalities.

Items:

-6

-3

0

3

6

-6i

-3i

3i

6i

$-\frac{\pi}{2}$

$-\frac{\pi}{3}$

$-\frac{\pi}{4}$

$-\frac{\pi}{6}$

$\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{\pi}{2}$

<

>

≤

≥

Adapted with permission from UCLES, A Level, June 2013, Paper 4725, Question 6.

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Complex Numbers: De Moivre 4ii

Subject & topics: Maths Stage & difficulty: Further A P2

Part A

$\cos^6 \theta$

By expressing $\cos \theta$ in terms of $e^{i\theta}$, write $\cos^6 \theta$ in terms of $\cos(n\theta)$.

Give your answer in the form

$$\cos^6 \theta = f(\cos 6\theta, \cos 4\theta, \cos 2\theta)$$

The following symbols may be useful: $\cos()$, θ

Part B

Solutions to an equation

Hence solve, for $0 \leq \theta \leq \pi$,

$$\cos 6\theta + 6 \cos 4\theta + 2 \cos 2\theta = 3.$$

Give your solutions in radians to 3 significant figures.

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Algebra and Roots: Cubics with Substitution 3i

Subject & topics: Maths **Stage & difficulty:** Further A P2

The cubic equation $x^3 + 2x^2 + 3x + 4 = 0$ has roots α , β and γ .

Part A

Substitution

Use the substitution $x = \frac{1}{u+1}$ to find a cubic equation in u with the form $au^3 + bu^2 + cu + d = 0$ where a, b, c and d are integers.

The following symbols may be useful: u

Part B

$$\left(\frac{1}{\alpha} - 1\right) \left(\frac{1}{\beta} - 1\right) \left(\frac{1}{\gamma} - 1\right)$$

Hence, find the value of $\left(\frac{1}{\alpha} - 1\right) \left(\frac{1}{\beta} - 1\right) \left(\frac{1}{\gamma} - 1\right)$ as a single fraction.

Adapted with permission from UCLES, A Level, June 2018, Paper 4725, Question 5.

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Matrices - Linear Equations 1

Subject & topics: Maths | Algebra | Matrices Stage & difficulty: Further A P3, University P2

Use matrix notation to solve the following set of three equations:

$$\begin{aligned}x + ky - z &= 3 \\ 3x + ky &= 1 \\ -x + 4y + z &= 3.\end{aligned}$$

where k is a constant.

Part A

Matrix form

Write these equations in matrix form $\mathbf{Ax} = \mathbf{b}$.

x

y

z

Part B

No unique solution

Find an expression for the condition that there is no unique solution to this set of equations.

The following symbols may be useful: k

Part C

The inverse matrix

Find the inverse matrix $\mathbf{A}^{-1}$.

$$\mathbf{A}^{-1} = \frac{1}{\boxed{}} \begin{pmatrix} k & \boxed{} & \boxed{} \\ \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & \boxed{} & \boxed{} \end{pmatrix}$$

Items:

- 4

-3

-1

0

1

3

4

-3k

-2k

-k

k

2k

3k

-k-12

-k-4

k+4

k+12

-3k-12

-3k-4

3k+4

3k+12

Part D

Solution to the set of equations

Using $\mathbf{A}^{-1}$, find the solutions for x , y and z in terms of k .

$$\left(\begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix}, \begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix}, \begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix} \right)$$

Items:

- 6

-4

-3

0

3

4

6

k+4

k+12

k-32

3k+4

3k+12

3k-32

4k+4

4k+12

4k-32

4-5k

12-5k

-5k-32



Matrices - Transformations 2

Subject & topics: Maths | Algebra | Matrices Stage & difficulty: Further A P1, University P1

A, **B** and **C** are 3×3 matrices such that $\mathbf{C} = \mathbf{BA}$ and

$$\mathbf{B} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$\mathbf{C} = \begin{pmatrix} p & 0 & q \\ 0 & r & 0 \\ s & 0 & t \end{pmatrix}$$

Part A
Matrix $\mathbf{B}^{-1}$

Find $\mathbf{B}^{-1}$.

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Part B

Matrix **A**

Use **B**^{−1} to find the matrix **A**.

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Part C

Transformation produced by **A**

Given that the matrix **A** represents rotation anticlockwise about the *y*-axis through an angle *D*, complete the matrix **A** using the items below.

A =

(

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Items:

- −1

0

1

− sin *D*

sin *D*

− cos *D*

cos *D*

− tan *D*

tan *D*

Part D

Reflection in the $z = 0$ plane

Given that $\mathbf{C}$ represents reflection in the $z = 0$ plane, find $\mathbf{C}$.

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Part E

Angle D

Deduce the value of the angle D . Give your answer in radians and assume $0 \leq D < 2\pi$.

The following symbols may be useful: pi

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Polar Coordinates: General 3i

Subject & topics: Maths **Stage & difficulty:** Further A P2

The equation of a curve, in polar coordinates, is

$$r = \sqrt{3} + \tan \theta, \quad \text{for } -\frac{1}{3}\pi \leq \theta \leq \frac{1}{4}\pi.$$

Part A

Tangent at the pole

Find the equation of the tangent at the pole in the form $\theta = \alpha$.

The following symbols may be useful: π , θ



Part B

Greatest value of r

State the greatest value of r .

Part C**Corresponding value of θ**

State the value of θ at which r takes its greatest value.

The following symbols may be useful: π

Part D

Sketch the curve

Sketch the curve.

Which curve in **Figure 1** most resembles your sketch?

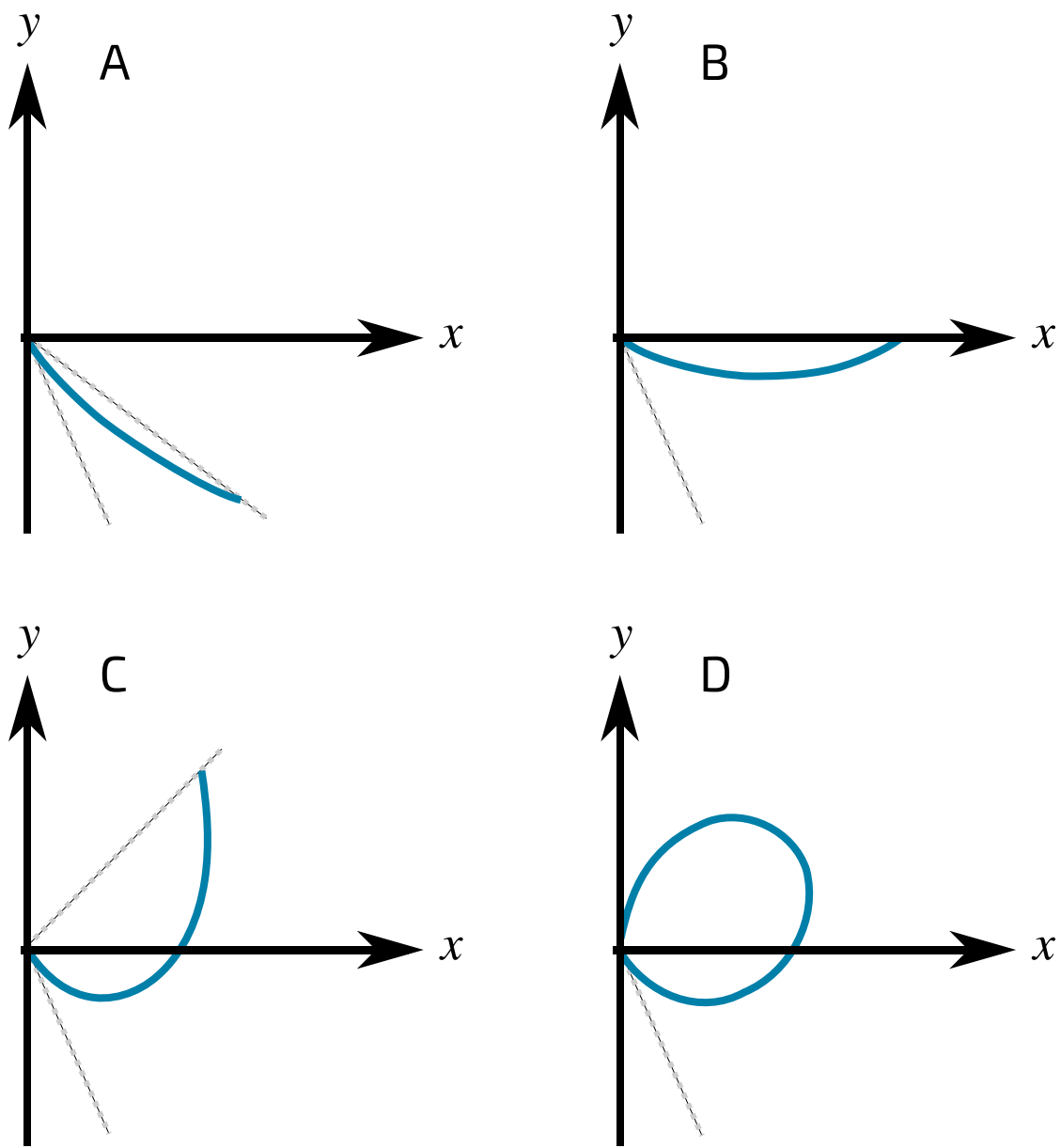


Figure 1: Four curves.

- ☐ Curve A
- ☐ Curve B
- ☐ Curve C
- ☐ Curve D

Part E

Area of region

Given that

$$\int \tan x \, dx = \ln |\sec x| + C,$$

find the exact area of the region enclosed by the curve and the lines $\theta = 0$ and $\theta = \frac{1}{4}\pi$.

The following symbols may be useful: `ln()`, `log()`, `pi`

Adapted with permission from UCLES, A Level, June 2006, Paper 4726, Question 7.

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Vectors: Lines and Planes 3i

Subject & topics: Maths **Stage & difficulty:** Further A P2

The plane Π passes through the points $(1, 2, 1)$, $(2, 3, 6)$ and $(4, -1, 2)$.

Part A

Cartesian equation of Π

Find a Cartesian equation of the plane Π .

Give your answer in the form $ax + by + cz = 19$.

The following symbols may be useful: x , y , z

Part B

Intersection of l and Π

The line l has equation $r = \begin{pmatrix} -1 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}$.

Find the value of λ at the point of intersection of Π and l .

Part C

Angle between l and Π

Find the acute angle between Π and l .

Give your answer in degrees to 3 significant figures.

Adapted with permission from UCLES, A Level, June 2015, Paper 4727, Question 3.

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Vectors: Geometry 2i

Subject & topics: Maths **Stage & difficulty:** Further A P2

(In this question, the notation $\triangle ABC$ denotes the area of the triangle ABC.)

The points P, Q and R have position vectors $p\mathbf{i}$, $q\mathbf{j}$ and $r\mathbf{k}$ respectively, relative to the origin O, where p , q and r are positive. The points O, P, Q and R are joined to form a tetrahedron.

Part A

Sketch tetrahedron

Draw a sketch of the tetrahedron.

Which of these sketches is correct?

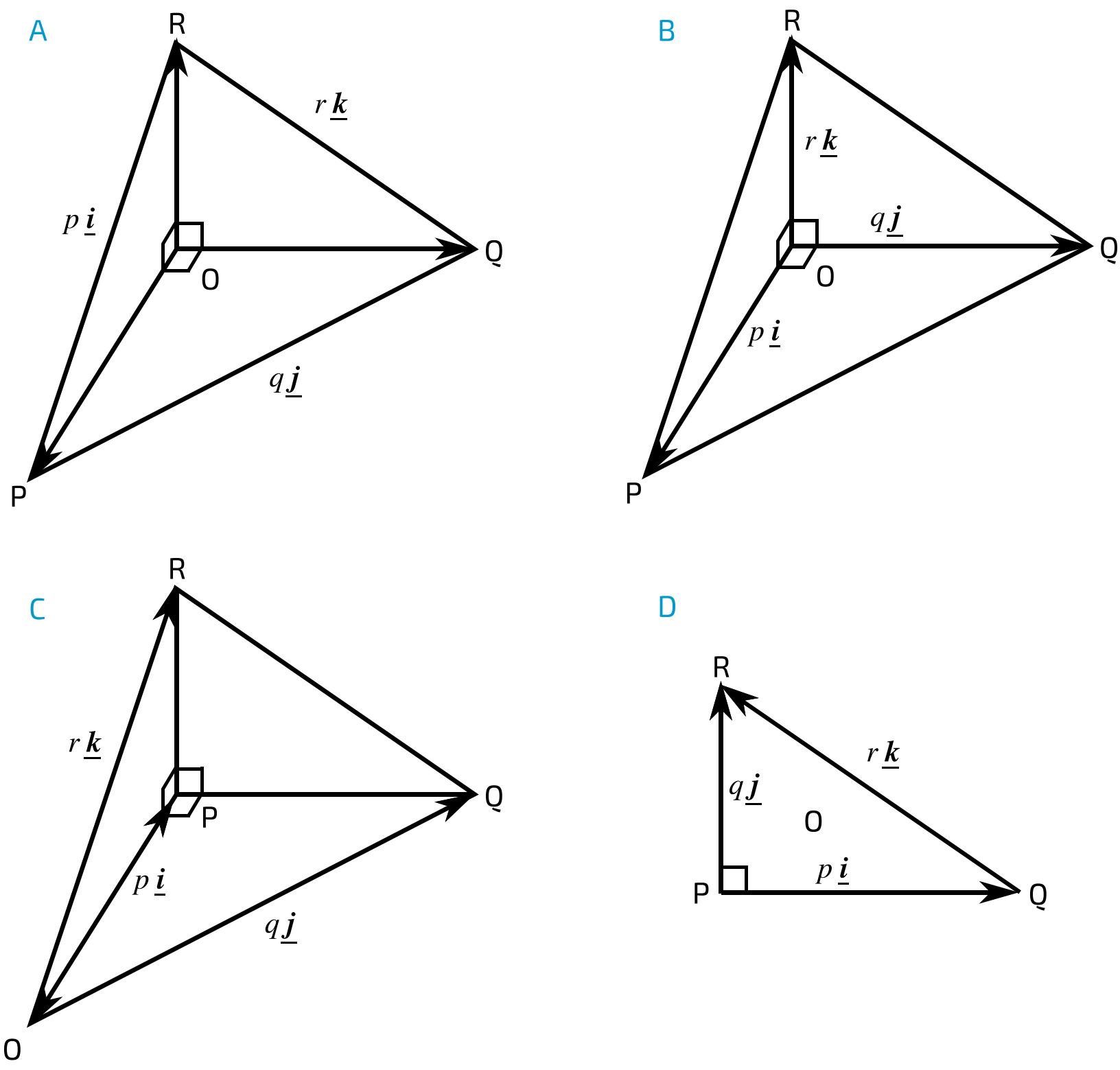


Figure 1: Four sketches.

- ☐ Sketch A
- ☐ Sketch B
- ☐ Sketch C
- ☐ Sketch D

Part B

Triangle areas

Find the values of $\triangle OPQ$, $\triangle OQR$ and $\triangle ORP$.

$\triangle OPQ =$

$\triangle OQR =$

$\triangle ORP =$

Items:

- $\frac{1}{2}qr$
- pqr
- $\frac{1}{2}pqr$
- $\frac{1}{3}qr$
- $\frac{1}{2}pr$
- pq
- $\frac{1}{3}pqr$
- qr
- $\frac{1}{3}pr$
- $\frac{1}{3}pq$
- $\frac{1}{2}pq$
- pr

Part C

Vector product and area

Use the definition of the vector product to show that $\frac{1}{2}|\overrightarrow{RP} \times \overrightarrow{RQ}|$ is equal to the area of one of the tetrahedron's faces.

Let $\angle PRQ = \theta$ and let $\hat{n}$ be a unit vector perpendicular to $\overrightarrow{RP}$ and $\overrightarrow{RQ}$. Then, we find that

$$\begin{aligned} \frac{1}{2}|\overrightarrow{RP} \times \overrightarrow{RQ}| &= \frac{1}{2}|\overrightarrow{RP}||\overrightarrow{RQ}||\text{}||\hat{n}| \\ &= \frac{1}{2}|\overrightarrow{RP}||\overrightarrow{RQ}|\text{} \\ &= \text{} \end{aligned}$$

Items:

- $\frac{1}{2}\triangle OPQ$
- $\frac{1}{2}\triangle OQR$
- $\frac{1}{2}\triangle OPR$
- $\frac{1}{2}\triangle PQR$
- $\triangle OPR$
- $\triangle OPQ$
- $\triangle OQR$
- $\triangle PQR$
- $2\triangle OPQ$
- $2\triangle OPR$
- $2\triangle OQR$
- $2\triangle PQR$
- $\sin \theta$
- $-\sin \theta$
- $\cos \theta$
- $-\cos \theta$

Part D

Relationship between areas

Show that we can find an equation of the form

$$(\triangle OPQ)^2 + (\triangle OQR)^2 + (\triangle ORP)^2 = \alpha(\triangle PQR)^2$$

where α is a constant to be found.

What is α ?

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